

Beyond Accuracy

A Deep Investigation into Failure Mechanisms of Classification Models

Project Report

Machine Learning

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Abstract

The adoption of machine learning models is accelerating in critical and sensitive sectors such as healthcare, law, and finance; however, the metric of “accuracy” alone is no longer sufficient to truly assess the effectiveness of these models. This report examines six major patterns of failure that may be masked by high accuracy metrics: the Baseline Illusion, Data Leakage, Classification Noise, Memorization, Vulnerability to Adversarial Attacks, and Algorithmic Bias. To deconstruct these issues, the report draws on a practical case study in the field of disease prediction, where analysis revealed 4,616 duplicate records out of 4,920, leading to contamination of the training and testing environments and artificially inflating the model’s accuracy to 100%. This study concludes that building reliable machine learning systems requires moving beyond evaluation using single metrics and shifting toward a comprehensive evaluation framework that includes data integrity checks, benchmark comparisons, and scrutiny of fairness and equity levels.

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Introduction

Machine learning models currently play a pivotal role in critical decision-making across multiple vital sectors, including healthcare, legal systems, financial services, and employment. With this growing influence, there is an urgent need to reevaluate the methodologies used to assess these models; while “accuracy” is commonly relied upon as a key performance indicator, it often provides an incomplete picture that does not reflect the system’s true efficiency.

A model may record high accuracy of up to 99%, yet suffer from fundamental structural flaws—such as memorizing training data rather than understanding it, relying on leaked data, or reproducing historical biases. In all these cases, numerical metrics remain high while the model fails in practice on the most critical tests.

This report aims to explore six hidden failure patterns that traditional accuracy metrics fail to detect, namely: the Baseline Illusion, Data Leakage, Label Noise, Memorization, Adversarial Vulnerability, and Algorithmic Bias. The report is accompanied by a practical case study in the field of disease prediction to illustrate how these failures manifest within granular machine learning pipelines and to reveal the critical consequences of relying on a single evaluation metric.

1.1 Why Accuracy Alone Is Misleading?

In fact, artificial intelligence can become “artificial stupidity” in some cases. Although the goal of developing artificial intelligence is to create technological systems that are intelligent and interact intelligently with their environment and users, they can sometimes exhibit behaviors or produce results that do not reflect true intelligence.

In cases of “artificial stupidity,” the intelligent system may have design or programming flaws that lead to incomprehensible or undesirable behavior. For example, the system may have difficulty understanding

instructions correctly, fail to perform a proper analysis of the input data, or even produce incorrect results due to programming glitches.

This highlights the importance of carefully developing and testing intelligent systems to ensure they behave correctly and effectively according to required standards, and are capable of effectively handling changing conditions in the environment.

1.2 Project Objectives

This project aims to:

1. Explain why accuracy alone is not sufficient for evaluating artificial intelligence systems.
2. Explain the concept of adversarial attacks and misleading examples.
3. Discuss the reasons why some models are susceptible to deception.
4. Review the real-world risks resulting from these attacks.
5. Emphasize the importance of building more secure and reliable artificial intelligence systems.
6. Demonstrate practical case study: Implement a disease prediction model in

Google Colab using the Kaggle Symptom-Disease dataset to show how data leakage can inflate accuracy metrics from realistic levels to artificially perfect results.

The Baseline Illusion — When a Dumb Model Wins

Machine learning models are frequently assessed by their precision; nonetheless, attaining a high accuracy level doesn't necessarily indicate that the model is smart or truly effective. In some cases, a simple model may perform the required task without actually understanding the underlying patterns within the data. This phenomenon is known as the “Baseline Illusion.” Therefore, comparing machine learning models with a baseline model is an essential step, as it helps determine whether the results truly reflect the model’s intelligence or are merely due to chance or data imbalance. We must ensure that the model performs better than the level of simple random guessing. Although accuracy scores are commonly used to evaluate model performance, a basic model may sometimes achieve similar results without truly benefiting from the available data. Therefore, comparing a model’s performance with a baseline model provides a clearer understanding of its actual effectiveness and reveals whether its performance is based on genuine superiority or merely the effect of chance.

2.1 What Is a Dummy Classifier?

The Dummy Classifier is a simple thing that does not need to learn from data to work. It just uses some ideas to make predictions like picking the most common thing it sees, class or randomly distributing predictions based on class proportions. The main purpose of a Dummy Classifier is to provide a baseline for comparing the performance of more advanced machine learning models. For example, if the majority of the data belongs to a specific class, the Dummy Classifier will predict that class for all input data. Although this may sometimes result in relatively high accuracy, such accuracy can be misleading because it does not reflect the model’s actual performance or its ability to understand patterns in the data and produce meaningful predictions.

2.2 When Simple Beats Complex

Sometimes, simple models are more effective than complex ones for several reasons. Complex models often require large amounts of data to be trained properly, but this can make them more prone to overfitting the training data, which negatively affects their ability to handle new data. In contrast, simple models are easier to train, faster to apply, and highly efficient when working with small or less complicated datasets. Increasing model complexity doesn't always mean performance. The key is picking a model that matches the data and the problem you're trying to solve. This way you get the results. Choosing the model is crucial. It should fit the data and the problem. That is how you achieve results, with your model.

2.3 Why Do You Always Need Baseline?

A baseline represents a fundamental element in evaluating the performance of machine learning models, as it plays a key role in determining how well a model performs. It helps verify whether the model is truly capable of learning and extracting patterns from real data or if it is merely relying on random guessing. Without a baseline, it becomes difficult to determine whether the model's performance reflects genuine learning or is simply due to chance. Therefore, baseline serves as an important reference point for measuring model efficiency and assessing its ability to accomplish the required tasks effectively. In addition, using a Dummy Classifier provides a useful way to compare model performance with simple prediction methods. If an advanced model fails to outperform the baseline, this may indicate that the model does not add real value or does not understand the data sufficiently. Therefore, baselines are considered essential for ensuring accurate model evaluation and improving their overall quality.

Data Leakage – The Silent Cheat

To ground the theoretical concepts in this report, a practical case study was conducted using a disease prediction model built in Google Colab. The study trained KNN, SVM, and Random Forest classifiers to predict 41 diseases from patient symptoms. The results serve as a direct illustration of the failure modes discussed in the following sections.

3.1 Types of Leakage

In the field of machine learning, data leakage is one of the most common patterns that easily misleads developers, not because it is difficult to understand, but because it appears as perfect numbers and results. This leads the programmer to believe they have achieved excellent results, when in reality the model has completely failed to understand the patterns.

This often occurs when the model is exposed to data it was never supposed to see.

There are two main types of data leakage:

One of the most obvious types is target leakage, where the leakage occurs indirectly when the data contains a clear hint of the final or expected answer from the model. For example, if we wanted to build a model to predict whether a person has diabetes, and the data provided to this model for training contained information indicating that the person is taking diabetes medication, the model would naturally give a correct prediction and impressive results. In reality, however, it has learnt nothing and has not produced the results based on an understanding of the relationship between the features, their analysis, and subsequently predicting the most likely outcome. The answer has simply been revealed to it through conclusive evidence of the disease, namely the presence of the medication. This means it will fail and become completely unable to make predictions when faced with new data.

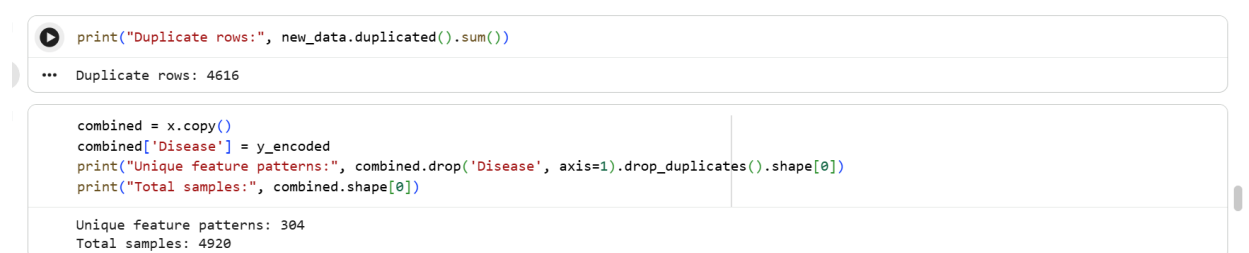
The other type is known as ‘train-test contamination’ and is considered more accurate. This type of data leakage occurs when the training set is accidentally or indirectly mixed with the test set during data preprocessing. One of the most common scenarios involving this type—and a prime example—is when we perform computations on the entire dataset before splitting it. What happens here is that the model learns statistical information about the test data that was supposed to be confidential, which explains the emergence of strong and ideal results whilst the model has not, in fact, learnt correctly.

The preprocessing phase consisted of the following steps:

1. Data Loading: 4,920 records loaded from Kaggle dataset with 18 columns
(1 Disease column + 17 Symptom columns)
2. Missing Value Analysis:
 - Symptom_4: 348 missing values (7.1%)
 - Symptom_5: 1,206 missing values (24.5%)
 - Symptom_17: 4,848 missing values (98.5%)
3. Symptom-to-Feature Conversion: 131 unique symptoms extracted and binarized
4. Duplicate Detection: Applied `.duplicated()` function to identify identical records

This problem was not merely theoretical; it emerged clearly in this study on ‘disease prediction’. We discovered that 4,616 out of 4,920 records were identical duplicates. This resulted in the same symptoms appearing in both the training and test sets. This is what enabled the model to achieve the highest possible accuracy of 100%, which is not usually possible, particularly in the field of medical diagnosis, but simply because it had already obtained the correct answers during the training process.

To ground this concept in a real example, the following output from our case study (Disease Prediction) directly illustrates how train-test contamination occurred:



```
print("Duplicate rows:", new_data.duplicated().sum())  
... Duplicate rows: 4616  
  
combined = x.copy()  
combined['Disease'] = y_encoded  
print("Unique feature patterns:", combined.drop('Disease', axis=1).drop_duplicates().shape[0])  
print("Total samples:", combined.shape[0])  
  
Unique feature patterns: 304  
Total samples: 4920
```

Figure 1: Duplicate records detected 4,616 out of 4,920 samples, with only 304 unique patterns.

3.2 How do data leaks deceive us with false accuracy?

Any programmer who starts building a model does so with the primary aim of achieving high accuracy, on the assumption that a high score indicates we have built a smart model capable of predicting or inferring perfectly. However, the results are often misleading and unrealistic due to data leakage. This is because, when bias occurs, the model has not understood the true relationships between inputs and outputs; rather,

it discovers shortcuts to reach the answer or may simply have memorised the answer. Consequently, when faced with new data, it fails completely, and the accuracy of the illusory figures is exposed.

How We Discovered the Problem:

The duplicate detection was performed using pandas `.duplicated()` function after completing all preprocessing steps. The analysis revealed:

- Original dataset: 4,920 records
- Duplicate records: 4,616 records (93.8%)
- Remaining unique patterns: 304 records (6.2%)

This massive overlap meant that when train-test split was performed (80-20), the same patient data appeared in both training set (243 samples) and testing set (61 samples), completely invalidating the model evaluation.

What makes this truly dangerous is the confidence this accuracy instils in programmers and decision-makers; when a model's accuracy reaches 99%, it becomes trusted, resources are allocated based on its predictions, and no one questions it. The biggest problem is also that the flaw only becomes apparent when it is too late – that is, after the model has been implemented – by which time the cost will be high.

The data leakage in our study manifested as follows:

- Initial Dataset Size: 4,920 patient records
- Duplicate Records Detected: 4,616 (93.8% of total data)
- Unique Feature Patterns: 304 (6.2% of total data)
- Symptom Features: 131 unique symptoms across all patients
- Disease Classes: 41 different diseases
- Top 15 Most Frequent Symptoms: fatigue, vomiting, high fever, loss of appetite, nausea, headache, abdominal pain, yellowish skin, yellowing of eyes, chills, skin rash, malaise, chest pain, joint pain, itching

This meant that 93.8% of the training data was repeated in the test set, allowing the model to memorize rather than learn.

This is what our study, “Disease Prediction”, revealed to us; once the data had been cleaned and the duplicate records that caused the data leakage had been removed, the dataset shrank dramatically from 4,920 records to just 304! This makes it medically insufficient to represent the complexity of patients in reality, even though the accuracy of the remaining data remained high. This proves to us that the model was not as robust as we initially thought; rather, it had not been tested against the real and unexpected variability found in the real world.

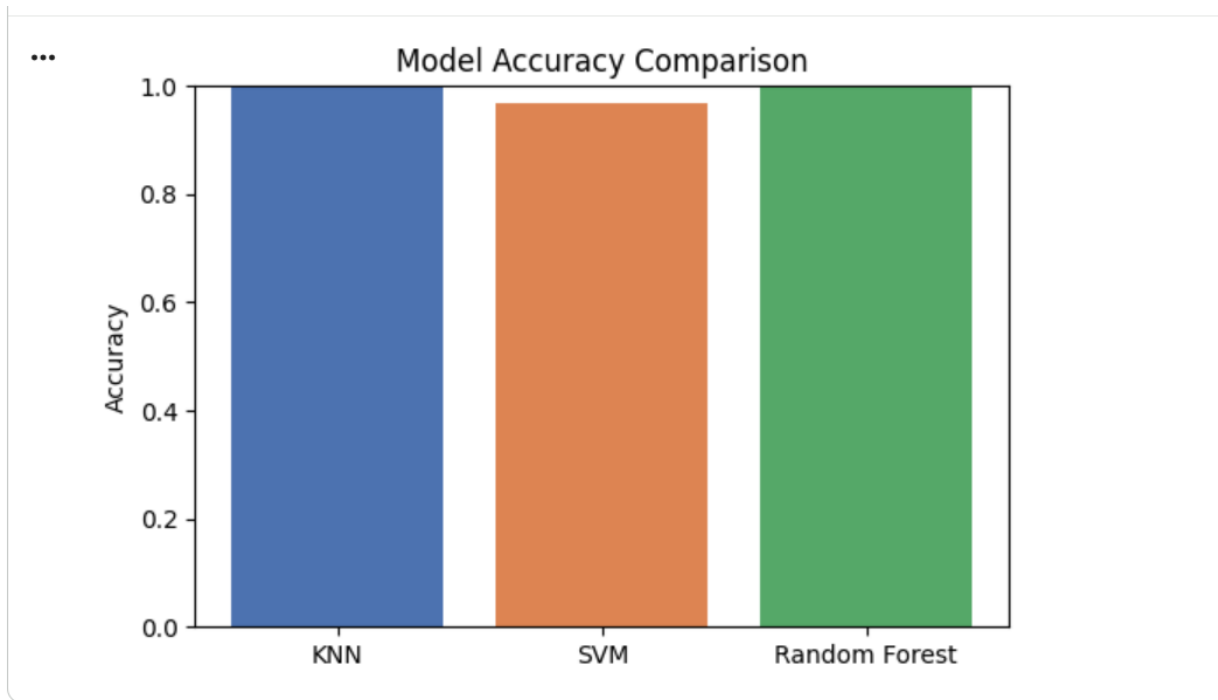


Figure 2: Model accuracy comparison after removing duplicate records.

The confusion matrix further confirms this - perfect diagonal predictions indicate the model memorized patterns rather than learned them:

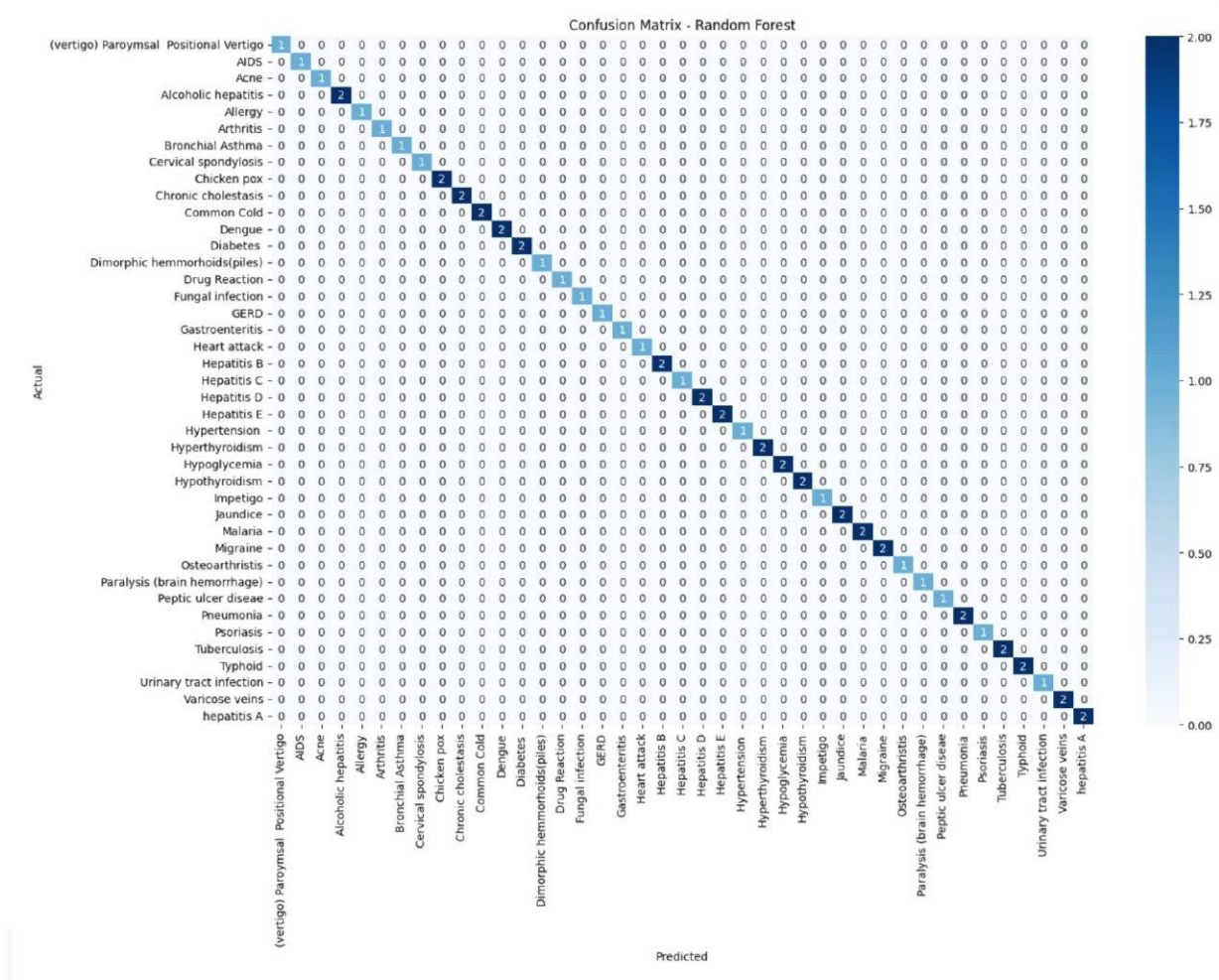


Figure 3: Confusion matrix of Random Forest - perfect diagonal across 41 disease classes.

Herein lies the real danger and cost behind that high accuracy and those impressive figures; illusory figures not only inflate performance results but also delay the discovery of fundamental flaws in the model. This delay in identifying the problem until the model has actually been implemented makes the consequences and risks far greater than they were at the outset.

3.3 Real-World Consequences

The consequences of data leakage are rarely detected within laboratories or controlled research environments; rather, the real shock becomes apparent during the deployment and actual operation phases. This is when it becomes impossible to ignore the vast gap between ideal results and the model's modest or failed performance during actual operation.

The severity of this problem was highlighted in a systematic review published in *Nature Machine Intelligence* by Roberts et al. (2021)

They examined 415 machine learning models built to diagnose COVID-19 patients using X-ray images. The result was unexpected: all models failed, and not a single one proved suitable for real-world application. The researchers later found that most of the studies contained a dataset known as “Frankenstein datasets,” which are data randomly collected from multiple public sources. This led to the same data and patient images appearing in both the training and test sets without the researchers’ knowledge. This type of data and overlap allowed the model to see the test set in advance, resulting in the loss of the results’ scientific credibility.

It wasn’t just a matter of fake accuracy figures; there was a complete shift in the model’s objective. In one experiment, X-ray images of children with pneumonia were compared with those of adults with COVID-19; the goal was to teach the model to diagnose the disease, but instead, the model learned the difference between children’s and adults’ lungs! Although its accuracy appeared strong and high, the primary objective of the task had completely shifted without the programmers noticing!

The danger here is that the flaw never surfaced during the traditional evaluation phase, and the models passed their tests successfully. But when researchers began trying to understand the logic behind its decisions and where it derived its information, the truth became clear.

Label Noise – Garbage In, Garbage Out

Data quality is considered a fundamental factor in determining the efficiency of machine learning models. When the training data has labels the models can learn the wrong things. This means the models will make predictions that’re not correct. The training data is very important because it teaches the models what to do. If the training data is bad then the models will be bad too. This is a problem because we need the models to make good predictions. If the models learn from training data they will make predictions that are not reliable. The training data needs to have labels so the models can learn the right things and make good predictions. This phenomenon is known as Label Noise. In addition, poor data quality can reduce the reliability of the model and negatively impact overall performance, whether the systems are used for classification or regression tasks.

4.1 What Is Label Noise?

Label Noise refers to the presence of errors or inaccuracies in data labels within a dataset. This problem occurs when data samples are assigned to incorrect categories due to factors such as human mistakes, system failures, or poor data collection methods. The presence of such noise can cause machine learning models to learn When we look at pictures sometimes the computer gets it wrong. This is a problem because it makes the computer less good at what it does. For example the computer might look at a picture of a cat. Say that it is a picture of a dog. This is a pattern. The computer is supposed to know what a cat looks like and what a dog looks like.. Sometimes it gets them mixed up. This is a mistake that affects the computers effectiveness. The computer is not good, at recognizing pictures of cats and dogs if it thinks a cat is a dog.

Pictures of cats are supposed to be labeled as cats, not dogs. If a cat may be incorrectly labeled as a dog in an image classification dataset. Such errors can confuse the model and lead to lower prediction accuracy. For this reason, data quality, cleanliness, and accurate labeling are considered essential factors in ensuring the development of reliable and effective machine learning systems.

4.2 Causes of Label Noise

Label noise can occur for several reasons, including human mistakes during data labeling, unclear data, or automatic labeling systems with low accuracy. In some cases, labels may also become incorrect due to data duplication or communication errors during data collection. These issues reduce data quality and make it more difficult for machine learning models to identify correct patterns. As a result, model performance and reliability may decrease.

4.3 Effects of Label Noise on Models

Label Noise represents a major challenge for machine learning models, as it leads to lower accuracy and increases the likelihood of prediction errors. When models are trained using inaccurate labels, they fail to identify the true underlying patterns and may instead learn incorrect relationships that provide little or no meaningful value. Furthermore, noisy labels can cause the problem of overfitting, where the model tends to memorize the incorrect information present in the training data rather than generalizing properly. As a result, the model's performance declines when it is applied to new data.

Memorization vs. Learning

Machine learning models are intended to understand patterns from data in order to achieve accurate predictions. However, in some cases, models may exhibit behavior that relies on memorizing the training data rather than learning the meaningful relationships and patterns within it. This thing can hurt how well the model works with data it hasn't seen before. So we need to tell the difference between memorizing and really learning. Memorization and true learning are important for the model's ability. The model needs to do learning instead of just memorizing. This is key, to handling unfamiliar data efficiently. Evaluating model reliability and ensuring effective performance.

5.1 What Is Memorization?

Memorization in machine learning models refers to the process in which a model stores specific examples from the training data instead of discovering the general underlying patterns within the data. In such cases, the model demonstrates excellent performance when dealing with the training data but performs poorly

when tested on new or unfamiliar data. This problem usually happens when the model is too complicated or when the dataset is not big enough. The model then has a time generalizing things correctly and getting the right answers. The model is not good, at generalizing because it is too complicated or the dataset is too small.

In the Disease Prediction case study, evidence of memorization was clearly visible:

1. Accuracy Metrics: All three models (KNN, SVM, Random Forest) achieved perfect 100% accuracy on the test set, which is statistically improbable for a medical diagnosis task involving 41 different disease categories.
2. Confusion Matrix Analysis: The Random Forest model's confusion matrix displayed a perfect diagonal pattern across all 41 disease classes, with zero off-diagonal predictions. This perfect diagonal is a textbook indicator of memorization rather than learning.
3. F1 Scores: Both macro F1 score and weighted F1 score reached exactly 1.0, which, combined with perfect accuracy, strongly indicates that the model had memorized the training data rather than learning generalizable patterns.
4. Root Cause: The discovery of 4,616 duplicate records out of 4,920 total samples (93.8% duplication rate) confirmed that the same patient cases appeared in both training and test sets, allowing the model to achieve artificially inflated metrics through memorization.

This case study demonstrates that perfect accuracy scores should immediately trigger investigation, as they are rarely achievable in real-world classification problems with 41 distinct classes.

5.2 What Is True Learning?

True learning occurs when a model is able to effectively recognize meaningful patterns and relationships within the data. When the model is trained properly, it becomes capable of applying what it has learned to interpret new data and generate highly accurate predictions. Unlike memorization, true learning enhances the model's ability to generalize, making it more efficient and reliable when applied in real-world situations and contexts.

5.3 Effects of Memorization on Model Performance

Relying on memorization can have a negative impact on the performance of machine learning models, as the model becomes highly dependent on specific examples from the training dataset. This often leads to a problem. The model gets too good at fitting the training data. It becomes very good at learning the noise and random fluctuations in that data. As a result, the model does not generalize well to unseen data. The model is overfitting. It is, like trying to draw a picture with too many details. You might draw the noise and random lines. You miss the overall picture. The model overfits when it tries hard to fit every single data point. The model learns the noise of the underlying patterns. This is overfitting. achieves very high accuracy on the training data but shows clear weakness when its performance is tested on new data. As a result, the model may fail to adapt to previously unseen data, which limits its effectiveness in real-world environments.

For this reason, reducing dependence on memorization is considered essential for developing machine learning systems that are efficient and capable of effective generalization.

Adversarial Examples - One Pixel Changes Everything

6.1 What Are Adversarial Attacks?

Misleading Examples: Small Changes, Big Results Adversarial attacks pose a significant challenge to artificial intelligence systems. These attacks involve manipulating the data fed into neural networks in clever and innovative ways to confuse the system and influence its outputs. There are many types of adversarial attacks, such as white-hat and black-hat attacks, as well as targeted and untargeted attacks.

For example, in the physical realm, adversarial attacks manifest as the exploitation of environmental conditions and external factors to influence the performance of deep learning models. For instance, clothing with specific designs and bright colors can be used to manipulate and confuse image recognition systems. This may lead to misclassification of images or bypassing security systems such as facial recognition.

In some cases, these techniques can be used to deceive self-driving cars and robots by obscuring traffic signs or altering the visual designs of roadways. They can also be used to confuse automated license plate recognition systems to evade detection and verification.

Researchers are working to develop methods to counter these attacks, such as improving the distinction between real and fake data and applying secure and reliable learning techniques.

6.2 Why Are AI Models So Easily Misled?

The researcher (Alvanpour, 2022) reviewed the structural reasons that make AI models susceptible to being easily misled, explaining that the problem lies in the following considerations:

1. **Relying on Irrelevant Features:** The model sometimes tends to learn secondary characteristics unrelated to the core task (such as skin color, gender, age, lighting quality, or device type) rather than understanding the true relationship between inputs and outputs.
2. **Lack of Data Diversity:** Training the model on limited or homogeneous samples (such as individuals from a single background) leads to its failure when faced with different scenarios. In this regard, the study emphasizes that relying on small, non-diverse samples masks biases and weakens the model's ability to generalize.
3. **Differences Between the Real World and the Training Environment:** Models often achieve excellent laboratory results but fail in real-world scenarios due to noise, lighting variations, and

changes in user or device behavior. Through her study, the researcher demonstrated that models appear more “fair” in the lab, while their biases become evident in actual application.

4. **Concept Drift:** Due to changes in temporal patterns—such as individual behavior, health conditions, and environmental circumstances—failure to update the model makes its predictions less accurate and more prone to errors.
5. **Focusing Exclusively on Accuracy Metrics:** The researcher criticized the practice of evaluating models solely based on metrics such as accuracy and F1-score without testing performance on diverse datasets, noting that her study revealed that only 5% of research explicitly examines and evaluates fairness.

The researcher (Alvanpour, 2022) concludes that the susceptibility of these models to bias stems from the fact that they lack human “awareness” and instead rely on pattern simulation; if the data used is biased, non-diverse, or limited in size, it will inevitably fail to accurately reflect reality.

6.3 Real-World Risks

The study (Alvanpour, 2022) moved from the theoretical aspect to an examination of the real and tangible risks that biased or inaccurate models can pose to individuals, particularly in sensitive sectors:

1. **Errors in Pulse Oximeters:** The researcher noted that these devices were often tested on individuals with light skin, leading to larger margins of error when applied to those with dark skin. This could result in a failure to detect actual oxygen deficiency, delayed medical intervention, and an increased likelihood of health complications.
2. **Atrial Fibrillation Detection:** The researcher explained that “false negative” errors in these systems are extremely dangerous because they fail to diagnose a condition that could lead to fatal complications, while frequent false alarms cause unnecessary anxiety and psychological distress for the patient.
3. **Predicting Complications of Pancreatic Cancer Surgery:** If the model classifies a high-risk patient as low-risk, they may be denied necessary medical care. Conversely, if the model overestimates risk, the decision to operate may be delayed, causing the patient to miss their only available treatment opportunity.
4. **Tracking Opioid Use:** The study noted that errors in monitoring systems may deprive patients in need of necessary support and treatment, or may lead to unfair accusations and penalties based on false predictions in criminal contexts.
5. **Speech and Text Recognition:** The researcher revealed the existence of clear biases against non-English speakers in text input and speech recognition systems, which increases error rates and excludes large segments of the population from benefiting from these technologies.

Algorithmic Bias – The Model Is Racist and Doesn't Know It

7.1 How Bias Creeps into Data

Machine learning models are not inherently biased; rather, they become biased gradually and subtly through the quality of the data they are trained on. This leads us to focus on whether the data we have reflects an inherently unfair reality, rather than on whether the programmer built discriminatory rules into the model. But data is essentially just a reflection of past human decisions. For example, if a bank in the past decided to grant loans based on personal biases, the model—in its natural tendency to discover and reproduce patterns—will automatically learn to replicate this behavior, without realizing that it was unfair or biased from the start.

One of the most prominent and obvious forms of bias in data is what is known as “representation bias.” This type of bias occurs when a specific category of data within the training dataset has been ignored or underrepresented, which affects the model’s understanding of it due to a lack of sufficient information to comprehend it, and consequently, its predictions regarding that category become weak and inaccurate. For example, facial recognition systems that have been extensively trained on people with light skin tones; as a result, their performance declines when dealing with people with dark skin tones due to “digital ignorance” resulting from an imbalance in the data.

What is known as “classification bias” is the most insidious type. It arises when a model relies on evaluation classifications created by humans who have personal biases and preconceptions, causing the model to turn these human biases into hard-coded rules. For example, when recruitment algorithms are trained on past hiring decisions, they automatically adopt the personal biases of hiring managers, as the algorithm learns what the “ideal employee” looks like based on a biased view of the past—one that may bear no relation to actual competence.

7.2 High Accuracy on Unfair Predictions

There is a troubling gap between numerical accuracy and true fairness when a model achieves very high accuracy but is at the same time biased and unfair; traditional measures of accuracy cannot detect this flaw, and therein lies the problem.

For example, suppose a model used in the judicial system predicts the likelihood of recidivism among previously convicted defendants. If the training data was drawn from a criminal justice system that has historically focused its surveillance disproportionately on certain communities or neighborhoods, the model will fully internalize these patterns.

As a result, the model will consistently predict a high risk of criminal behavior among individuals from those communities, and this prediction will appear accurate and statistically validated according to the

model's own criteria—precisely because it reproduces the very bias and injustice inherent in the original data.

Here it becomes clear that accuracy may be as real as the numbers themselves, but it completely lacks fairness and equity.

This was not merely a hypothetical scenario; the well-known case of the risk assessment tool (COMPAS) used in the United States has proven that it incorrectly classified Black offenders as “high risk” compared to white defendants, while misclassifying white defendants as “low risk” at varying rates.

What is surprising here is that the model's overall accuracy was nearly equal between the two groups, which made the system appear successful; however, the distribution and nature of the errors were by no means fair. This proves that the model's accuracy metric calculates only the overall percentage, without clarifying which group actually bears the real and costly burden of these errors.

7.3 Famous Real-World Cases

COMPAS — United States, 2016

One of the most prominent examples in research on the fairness of smart systems is the famous investigation conducted by ProPublica in 2016 into the COMPAS algorithm, which is designed to predict the likelihood of defendants reoffending. Its predictions carried real legal weight, as many U.S. states relied on this tool as a primary means of making critical decisions such as setting bail amounts, issuing judicial rulings, and granting parole. The real problem is that this grave error of racial bias committed by the system remained hidden and unexamined until journalists analyzed the data—but only after the algorithm had already affected the fates of thousands of defendants and their cases.

Amazon's Hiring Algorithm — 2018

Amazon developed a machine learning tool to sort through job applicants' resumes. The model was trained on the company's historical hiring data, which spanned a period when men were significantly more prevalent than women in the tech sector. As a result, the model learned this bias and began automatically filtering out resumes containing words like “feminine” and downgrading their ratings, as well as those from women-only colleges. Although Amazon scrapped the project before it was officially launched, it serves as an example of how bias in training data can directly translate into discriminatory outcomes against specific groups.

Healthcare Resource Allocation — 2019

In the medical field, a study published in the journal *Science* revealed a serious flaw in a model used by U.S. hospitals to allocate additional healthcare resources to the most at-risk patients. The algorithm measured a patient's need for care based on the patient's “previous financial expenditure.” This may seem like a logical assumption, but in reality, it is a bias. Historical data on Black patients shows that their healthcare spending was lower due to systemic and financial barriers that prevent them from accessing care, not because they are healthier. However, the algorithm interpreted their lower spending as a lack of need

for treatment. As a result of the consistent underestimation of these patients' health risks, more than 50% of them were denied the additional care they actually deserved.

Conclusion & Recommendations

8.1 Summary of Hidden Failure Modes

From this, we can conclude that the metrics we rely on to evaluate machine learning models do not always represent the reality we think we are measuring, and this is the idea that emerged and has been a recurring theme throughout this report. The accuracy metric operates in a misleading manner by imposing absolute confidence that the model may not actually deserve; structural flaws and hidden biases may be concealed behind these seemingly perfect numbers.

Although these vulnerabilities manifest in different ways, they pose the same risk: they pass through the evaluation phase undetected. For example, “data leakage” allows the model to see confidential data it is not supposed to see, thereby creating false accuracy; conversely, the “baseline illusion” provides ideal numbers to a model that has learned only to repeat the most frequent outcome. Meanwhile, “classification noise” corrupts the foundation of learning by teaching the model non-existent patterns, while “blind memorization” yields perfect accuracy without true understanding. On the other hand, “attack fragility” comes to destroy the model we thought was robust with just a simple manipulation of the inputs. Finally, “algorithmic bias” emerges when the model is trained on an unbalanced dataset; in such cases, the model appears to have achieved high accuracy when its performance is measured across the entire dataset, while it fails on the underrepresented subsets of the data.

The Disease Prediction case study served as a concrete example of how these theoretical failures manifest in practice. The dataset containing 4,920 disease-symptom records appeared to offer excellent classification results (100% accuracy across KNN, SVM, and Random Forest), yet 93.8% of the data consisted of exact duplicates. When cleaned and reduced to 304 unique patterns, the models still achieved perfect accuracy, indicating that memorization and data leakage had rendered the evaluation metrics completely unreliable.

The confusion matrix revealed a perfect diagonal pattern—a telltale sign that the models had learned to recognize specific training examples rather than discovering the underlying medical relationships between symptoms and diseases.

None of the failure cases discussed here were rare or exceptional; on the contrary, these errors appeared consistently in published scientific research, in software systems actually deployed, and even in models that won technical competitions. On this basis, the scope of actual application and the extent of its real-world adoption is the only difference between a model that fails quietly within a development and laboratory environment, and another model that fails catastrophically in a hospital or courtroom.

8.2 Practical Checklist Before Trusting Any Model

It is recommended to seek accurate and verified answers to the following questions before adopting the performance of any technical model and considering it reliable:

On the data:

- Was the train-test split performed before any preprocessing?
- Are there duplicate records that could have crossed the split boundary?
- Were the labels assigned by humans — and if so, by whom, under what conditions?
- Does the dataset represent all relevant groups fairly?

On the evaluation:

- Has the model been compared against a simple baseline?
- Were metrics beyond accuracy examined — F1, precision, recall, or MAE depending on the task?
- Were error rates analyzed separately across subgroups, not just in aggregate?
- Does the model's confidence align with its actual correctness?

On deployment:

- Will the features available at training time still be available at prediction time?
- Has the model been tested on data collected independently from its training set?
- Is there a mechanism to monitor performance after deployment, not just before?

Although checking this list does not guarantee the complete elimination of potential risks, systematically addressing these questions makes it much more difficult for the unnoticed failures described in this report to pass through the evaluation phase undetected.

8.3 Future Directions

As the use of machine learning models expands into medical, legal, financial, and recruitment systems, the severity of the issues documented in this report is increasing, and they continue to emerge. Consequently, the importance of developing mechanisms for detection and remediation is growing. Prominent current examples of solutions include explanation tools such as SHAP and LIME, which allow programmers to identify the features influencing the model's decision and thereby uncover vulnerabilities such as leakage and bias hidden behind seemingly perfect accuracy figures. Scientific research is also moving toward the

field of “fairness-aware learning” to develop methodologies aimed at improving the fair distribution of error rates across categories, rather than focusing solely on the model’s overall performance. In addition, automated data pipelines are being built to inspect data in production environments, a step that ensures the detection of any data contamination and prevents it from reaching active models.

Here, the shift must be twofold—in scientific culture and not just in the technical aspect—and this is crucial, as the field is gradually moving away from the idea of treating a single performance metric as the endpoint of research. The evaluation of intelligent systems no longer stops at the number of correct predictions but involves a deeper analysis, requiring an examination of the quality of errors, their distribution across categories, the causes of their occurrence, and the consequences resulting from system failure. Future models do not necessarily need to be more accurate than those currently available, but they are expected to be more accurate and objective in interpreting the reliability of their figures and what they actually mean in the operational environment.

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Appendix A: Practical Case Study

Title: Disease Prediction Based on Symptoms

Tool: Google Colaboratory

Dataset: Kaggle — Symptom-Disease Dataset (4,920 records, 41 diseases)

Colab Notebook:

<https://colab.research.google.com/drive/1levE6UzRFVgXDlkkdMgV6J9Ix83OYzIe?usp=sharing>

The notebook covers:

A. Required Libraries:

- pandas: Data manipulation and analysis
- numpy: Numerical computations
- matplotlib.pyplot: Data visualization
- seaborn: Statistical data visualization
- scikit-learn: Machine learning models and metrics
 - sklearn.preprocessing (MultiLabelBinarizer, LabelEncoder)
 - sklearn.model_selection (train_test_split)
 - sklearn.neighbors (KNeighborsClassifier)
 - sklearn.svm (SVC)
 - sklearn.ensemble (RandomForestClassifier)
 - sklearn.metrics (accuracy_score, classification_report, confusion_matrix, f1_score)

B. Dataset Statistics:

- Total Records: 4,920
- Unique Diseases: 41
- Unique Symptoms: 131
- Duplicate Records Found: 4,616
- Unique Feature Patterns: 304
- Train Set Size: 243 samples
- Test Set Size: 61 samples
- Test Size Percentage: 20% (test_size=0.2)

C. Model Parameters Used:

1. KNN Classifier: n_neighbors=5
2. SVM Classifier: kernel='linear', random_state=42
3. Random Forest Classifier: random_state=42

D. Data Preprocessing Steps:

- Missing values filled with empty strings
- Symptoms converted to binary features using MultiLabelBinarizer
- Disease labels encoded using LabelEncoder
- Train-test split with stratification (stratify=y_clean)
- Random state set to 42 for reproducibility

E. Results Summary:

- All three models achieved 100% accuracy on test set
- Macro F1 Score: 1.0
- Weighted F1 Score: 1.0

- Confusion matrix showed perfect diagonal predictions